

# Timing Analysis

## Parcial 2, Parte 1 — FSM Semáforo con Cruce Peatonal

José Rueda

Arquitectura de Computadoras — Universidad del Istmo de Guatemala

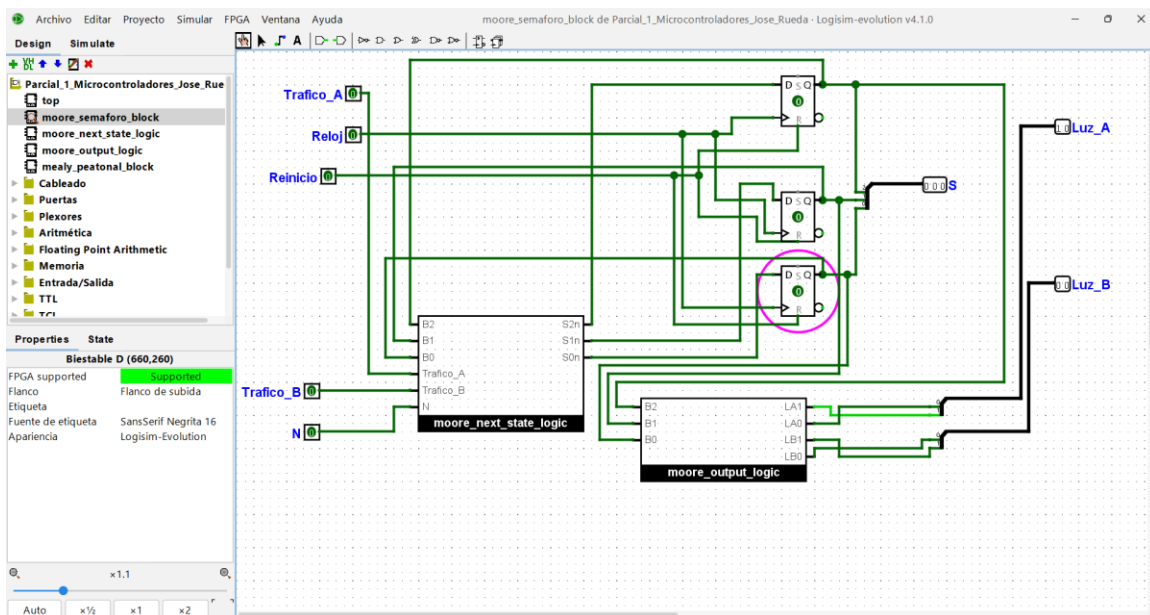
### 1. Objetivo

Este documento calcula, a partir de piezas lógicas reales disponibles en Digikey, la frecuencia máxima de operación y el cumplimiento de la restricción de hold para un tramo de flip-flop a flip-flop que cruza entre la máquina de estados Moore (moore\_semaforo\_block) y la máquina de estados Mealy (mealy\_peatonal\_block) del proyecto de la FSM del semáforo.

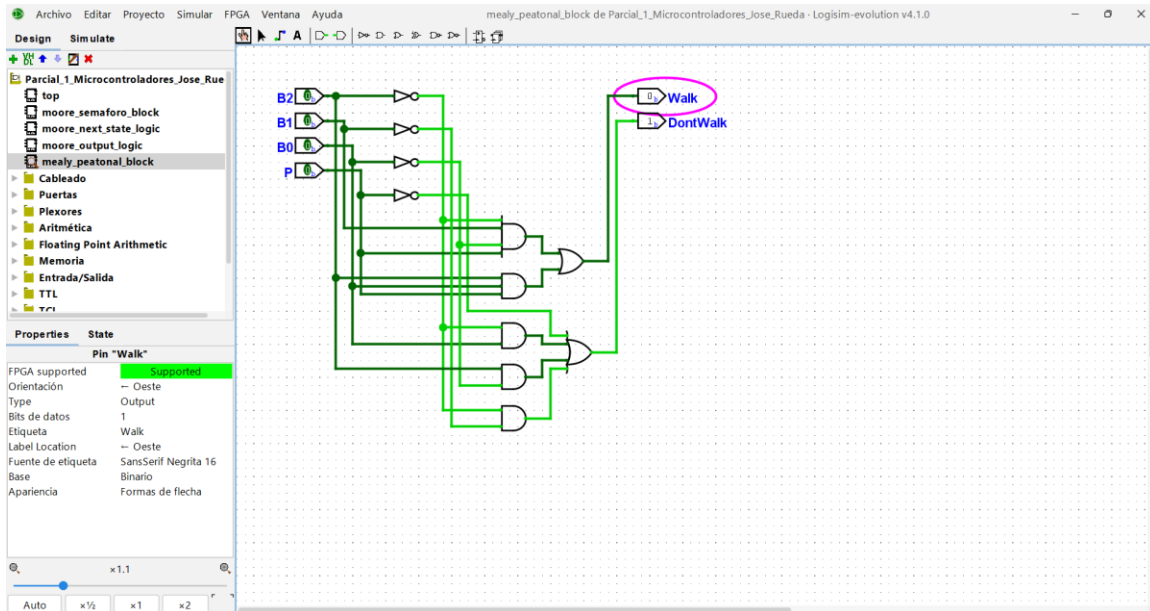
### 2. Definición del tramo

El tramo analizado comienza en el biestable D correspondiente al bit B0 del registro de estado de la FSM Moore, atraviesa la lógica combinacional de la FSM Mealy que calcula la señal Walk, y termina en un registro de captura hipotético a la salida de Walk (práctica estándar de diseño para evitar glitches en una salida que sale del sistema).

#### 2.1 FF de origen y lógica de próximo estado — moore\_semaforo\_block



## 2.2 Lógica combinacional de destino — señal Walk



Ecuación booleana minimizada implementada:

$$Walk = B0 \cdot B2 \cdot P + B1 \cdot P \cdot \sim B0 \cdot \sim B2$$

## 3. Selección de componentes reales (Digikey / Texas Instruments)

Familia lógica 74HC (CMOS de alta velocidad), disponible en Digikey. Los valores de las tablas siguientes corresponden a VCC=4.5V, sobre el rango de temperatura comercial completo (-40°C a 85°C).

### 3.1 SN74HC74N — Flip-flop D

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**SN74HC74N**

DigiKey Part Number: 296-1602-5-ND

Manufacturer: [Texas Instruments](#)

Manufacturer Product Number: SN74HC74N

Description: IC FF D-TYPE 2-ELE 1-BIT 14-PDIP

Manufacturer Standard Lead Time: 16 Weeks

Customer Reference:

Detailed Description: Flip Flop 2 Element D-Type 1 Bit Positive Edge 14-DIP (0.300", 7.62mm)

Datasheet: [Datasheet](#)

EDA/CAD Models: [SN74HC74N Models](#)

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| 10       | \$0.81500  | \$8.15    |
| 25       | \$0.73680  | \$18.42   |
| 100      | \$0.65070  | \$65.07   |
| 250      | \$0.60964  | \$152.41  |

Product Attributes

| Category | Integrated Circuits (ICs) | Logic | Max Propagation Delay @ V, Vmax | 30ns @ 6V, 50pF |
|----------|---------------------------|-------|---------------------------------|-----------------|
|          |                           |       |                                 |                 |

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## 6.9 Switching Characteristics - 74

over operating free-air temperature range (unless otherwise noted)

| PARAMETER        |                   | FROM       | TO                  | V <sub>CC</sub> | Operating free-air temperature (T <sub>A</sub> ) |     |     |               |     |     | UNIT |
|------------------|-------------------|------------|---------------------|-----------------|--------------------------------------------------|-----|-----|---------------|-----|-----|------|
|                  |                   |            |                     |                 | 25°C                                             |     |     | −40°C to 85°C |     |     |      |
|                  |                   |            |                     |                 | MIN                                              | TYP | MAX | MIN           | TYP | MAX |      |
| f <sub>max</sub> |                   |            |                     | 2 V             | 6                                                | 10  |     | 6             |     |     | MHz  |
|                  |                   |            |                     | 4.5 V           | 31                                               | 50  |     | 25            |     |     |      |
|                  |                   |            |                     | 6 V             | 36                                               | 60  |     | 29            |     |     |      |
| t <sub>pd</sub>  | Propagation delay | PRE or CLR | Q or $\overline{Q}$ | 2 V             |                                                  | 70  | 230 |               |     | 290 | ns   |
|                  |                   |            |                     | 4.5 V           |                                                  | 20  | 46  |               |     | 58  |      |
|                  |                   |            |                     | 6 V             |                                                  | 15  | 39  |               |     | 49  |      |
|                  |                   | CLK        | Q or $\overline{Q}$ | 2 V             |                                                  | 70  | 175 |               |     | 220 |      |
|                  |                   |            |                     | 4.5 V           |                                                  | 20  | 35  |               |     | 44  |      |
|                  |                   |            |                     | 6 V             |                                                  | 15  | 30  |               |     | 39  |      |
| t <sub>t</sub>   | Transition-time   |            | Q or $\overline{Q}$ | 2 V             |                                                  | 28  | 75  |               |     | 95  | ns   |
|                  |                   |            |                     | 4.5 V           |                                                  | 8   | 15  |               |     | 19  |      |
|                  |                   |            |                     | 6 V             |                                                  | 6   | 13  |               |     | 16  |      |

| Parámetro     | MIN   | TYP (25°C) | MAX   |
|---------------|-------|------------|-------|
| tCO (Clock→Q) | 20 ns | 35 ns      | 44 ns |
| tsu (setup)   | —     | —          | 25 ns |
| th (hold)     | —     | —          | 0 ns  |

### 3.2 SN74HC04N — Inversor

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**SN74HC04N**

**DigiKey Part Number** 296-1566-5-ND

**Manufacturer** [Texas Instruments](#)

**Manufacturer Product Number** SN74HC04N

**Description** IC INV 6-CH 1-IN 14-PDIP

**Manufacturer Standard Lead Time** 16 Weeks

**Customer Reference**

**Detailed Description** Inverter IC 6 Channel 14-PDIP

**EDA/CAD Models** [SN74HC04N Models](#)

Image shown is a representation only. Exact specifications should be obtained from the product data sheet.

**Product Attributes**

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| Category | <a href="#">Integrated Circuits (ICs)</a><br><a href="#">Logic</a><br><a href="#">Gates and Inverters</a> | Current - Output High, Low | 5.2mA, 5.2mA |
|----------|-----------------------------------------------------------------------------------------------------------|----------------------------|--------------|
| Mfr      | <a href="#">Texas Instruments</a>                                                                         | Input Logic Level - Low    | 0.5V ~ 1.8V  |

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| 1        | \$0.81000  | \$0.81    |
| 10       | \$0.57800  | \$5.78    |
| 25       | \$0.52000  | \$13.00   |
| 100      | \$0.45580  | \$45.58   |
| 250      | \$0.42512  | \$106.28  |

| Parámetro | MIN  | TYP (25°C) | MAX   |
|-----------|------|------------|-------|
| tpd (A→Y) | 9 ns | 19 ns      | 24 ns |

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[Texas Instruments SN74HC11N](#)

**SN74HC11N**

DigiKey Part Number: 296-8217-5-ND

Manufacturer: [Texas Instruments](#)

Manufacturer Product Number: SN74HC11N

Description: IC AND 3-CIR 3-IN 14-PDIP

Manufacturer Standard Lead Time: 9 Weeks

Customer Reference:

Detailed Description: AND Gate IC 3 Channel 14-PDIP

Datasheet: [Datasheet](#)

EDA/CAD Models: [SN74HC11N Models](#)

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| QUANTITY | UNIT PRICE | EXT PRICE |
|----------|------------|-----------|
| 1        | \$0.89000  | \$0.89    |
| 10       | \$0.63800  | \$6.38    |
| 25       | \$0.57400  | \$14.35   |
| 100      | \$0.50400  | \$50.40   |
| 250      | \$0.47064  | \$117.66  |

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|----------|-----------------------------------------------------------------------------------------------------------|----------------------------|--------------|

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over operating free-air temperature range (unless otherwise noted)

| PARAMETER       |                   | FROM       | TO | V <sub>CC</sub> | Operating free-air temperature (T <sub>A</sub> ) |     |     |               |     |     | UNIT |
|-----------------|-------------------|------------|----|-----------------|--------------------------------------------------|-----|-----|---------------|-----|-----|------|
|                 |                   |            |    |                 | 25°C                                             |     |     | –40°C to 85°C |     |     |      |
|                 |                   |            |    |                 | MIN                                              | TYP | MAX | MIN           | TYP | MAX |      |
| t <sub>pd</sub> | Propagation delay | A, B, or C | Y  | 2 V             | 35                                               |     | 100 | 125           |     |     | ns   |
|                 |                   |            |    | 4.5 V           | 10                                               |     | 20  | 25            |     |     |      |
|                 |                   |            |    | 6 V             | 8                                                |     | 17  | 21            |     |     |      |
| t <sub>t</sub>  | Transition-time   |            | Y  | 2 V             | 25                                               |     | 75  | 95            |     |     | ns   |
|                 |                   |            |    | 4.5 V           | 7                                                |     | 15  | 19            |     |     |      |
|                 |                   |            |    | 6 V             | 5                                                |     | 13  | 16            |     |     |      |

| Parámetro     | MIN   | TYP (25°C) | MAX   |
|---------------|-------|------------|-------|
| tpd (A,B,C→Y) | 10 ns | 20 ns      | 25 ns |

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**SN74HC21N**

DigiKey Part Number: 296-8266-5-ND

Manufacturer: [Texas Instruments](#)

Manufacturer Product Number: SN74HC21N

Description: IC AND 2-CIR 4-IN 14-PDIP

Manufacturer Standard Lead Time: 9 Weeks

Customer Reference:

Detailed Description: AND Gate IC 2 Channel 14-PDIP

Datasheet: [Datasheet](#)

EDA/CAD Models: [SN74HC21N Models](#)

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|----------|------------|-----------|
| 1        | \$1.04000  | \$1.04    |
| 10       | \$0.74800  | \$7.48    |
| 25       | \$0.67520  | \$16.88   |
| 100      | \$0.59470  | \$59.47   |
| 250      | \$0.55628  | \$139.07  |

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|----------------------------|--------------|
| Integrated Circuits (ICs)  | Logic        |
| Current - Output High, Low | 5.2mA, 5.2mA |

over operating free-air temperature range (unless otherwise noted)

| PARAMETER       |                   | FROM          | TO | V <sub>CC</sub> | Operating free-air temperature (T <sub>A</sub> ) |     |     |               |     |     | UNIT |
|-----------------|-------------------|---------------|----|-----------------|--------------------------------------------------|-----|-----|---------------|-----|-----|------|
|                 |                   |               |    |                 | 25°C                                             |     |     | –40°C to 85°C |     |     |      |
|                 |                   |               |    |                 | MIN                                              | TYP | MAX | MIN           | TYP | MAX |      |
| t <sub>pd</sub> | Propagation delay | A, B, C, or D | Y  | 2 V             |                                                  | 44  | 110 |               |     | 140 | ns   |
|                 |                   |               |    | 4.5 V           |                                                  | 14  | 22  |               |     | 28  |      |
|                 |                   |               |    | 6 V             |                                                  | 11  | 19  |               |     | 24  |      |
| t <sub>t</sub>  | Transition-time   |               | Y  | 2 V             |                                                  | 29  | 75  |               |     | 95  | ns   |
|                 |                   |               |    | 4.5 V           |                                                  | 10  | 15  |               |     | 19  |      |
|                 |                   |               |    | 6 V             |                                                  | 8   | 13  |               |     | 16  |      |

| Parámetro                   | MIN   | TYP (25°C) | MAX   |
|-----------------------------|-------|------------|-------|
| t <sub>pd</sub> (A,B,C,D→Y) | 14 ns | 22 ns      | 28 ns |

### 3.5 SN74HC32N — OR de 2 entradas

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**SN74HC32N**

DigiKey Part Number: 296-1589-5-ND

Manufacturer: [Texas Instruments](#)

Manufacturer Product Number: SN74HC32N

Description: IC OR 4-CH 2-IN 14-PDIP

Manufacturer Standard Lead Time: 9 Weeks

Customer Reference:

Detailed Description: OR Gate IC 4 Channel 14-PDIP

Datasheet: [Datasheet](#)

EDA/CAD Models: [SN74HC32N Models](#)

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| QUANTITY | UNIT PRICE | EXT PRICE |
|----------|------------|-----------|
| 1        | \$0.70000  | \$0.70    |
| 10       | \$0.49700  | \$4.97    |
| 25       | \$0.44680  | \$11.17   |
| 100      | \$0.39070  | \$39.07   |
| 250      | \$0.36408  | \$91.02   |

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| Category                   | Integrated Circuits (ICs) | Logic | Gates and Inverters |
|----------------------------|---------------------------|-------|---------------------|
| Current - Output High, Low | 5.2mA, 5.2mA              |       |                     |

### 6.8 Switching Characteristics - 74

$C_L = 50$  pF, over operating free-air temperature range (unless otherwise noted)

| PARAMETER       |                   | FROM   | TO | V <sub>CC</sub> | Operating free-air temperature (T <sub>A</sub> ) |     |     |               |     |     | UNIT |
|-----------------|-------------------|--------|----|-----------------|--------------------------------------------------|-----|-----|---------------|-----|-----|------|
|                 |                   |        |    |                 | 25°C                                             |     |     | −40°C to 85°C |     |     |      |
|                 |                   |        |    |                 | MIN                                              | TYP | MAX | MIN           | TYP | MAX |      |
| t <sub>pd</sub> | Propagation delay | A or B | Y  | 2 V             | 50                                               | 100 | 125 |               |     | ns  |      |
|                 |                   |        |    | 4.5 V           | 10                                               | 20  | 25  |               |     |     |      |
|                 |                   |        |    | 6 V             | 8                                                | 17  | 21  |               |     |     |      |

$C_L = 50$  pF, over operating free-air temperature range (unless otherwise noted)

| PARAMETER      |                 | FROM | TO | V <sub>CC</sub> | Operating free-air temperature (T <sub>A</sub> ) |     |     |               |     |     | UNIT |
|----------------|-----------------|------|----|-----------------|--------------------------------------------------|-----|-----|---------------|-----|-----|------|
|                |                 |      |    |                 | 25°C                                             |     |     | −40°C to 85°C |     |     |      |
|                |                 |      |    |                 | MIN                                              | TYP | MAX | MIN           | TYP | MAX |      |
| t <sub>t</sub> | Transition-time |      | Y  | 2 V             | 38                                               |     | 75  | 95            |     |     | ns   |
|                |                 |      |    | 4.5 V           | 8                                                |     | 15  | 19            |     |     |      |
|                |                 |      |    | 6 V             | 6                                                |     | 13  | 16            |     |     |      |

| Parámetro   | MIN   | TYP (25°C) | MAX   |
|-------------|-------|------------|-------|
| tpd (A,B→Y) | 10 ns | 20 ns      | 25 ns |

Fuente: hojas de datos oficiales de Texas Instruments — [ti.com/lit/ds/symlink/sn74hc74.pdf](https://www.ti.com/lit/ds/symlink/sn74hc74.pdf), [sn74hc04.pdf](https://www.ti.com/lit/ds/symlink/sn74hc04.pdf), [sn74hc11.pdf](https://www.ti.com/lit/ds/symlink/sn74hc11.pdf), [sn74hc21.pdf](https://www.ti.com/lit/ds/symlink/sn74hc21.pdf), [sn74hc32.pdf](https://www.ti.com/lit/ds/symlink/sn74hc32.pdf) (secciones "Timing Requirements" y "Switching Characteristics"), también disponibles en Digikey.

## 4. Identificación de la ruta crítica

La ecuación de Walk tiene dos términos producto, con distinta profundidad de compuertas:

- $B0 \cdot B2 \cdot P$ : pasa por una AND de 3 entradas  $\rightarrow$  OR de 2 entradas. 2 etapas.
- $B1 \cdot P \cdot \sim B0 \cdot \sim B2$ : requiere invertir  $B0$  y  $B2$  primero (NOT), luego una AND de 4 entradas, y finalmente la OR de 2 entradas. 3 etapas — la ruta más larga.

**Ruta crítica (máxima): NOT  $\rightarrow$  AND-4  $\rightarrow$  OR-2**

**Ruta más corta (para hold): AND-3  $\rightarrow$  OR-2**

## 5. Diagrama de secuencia de tiempo

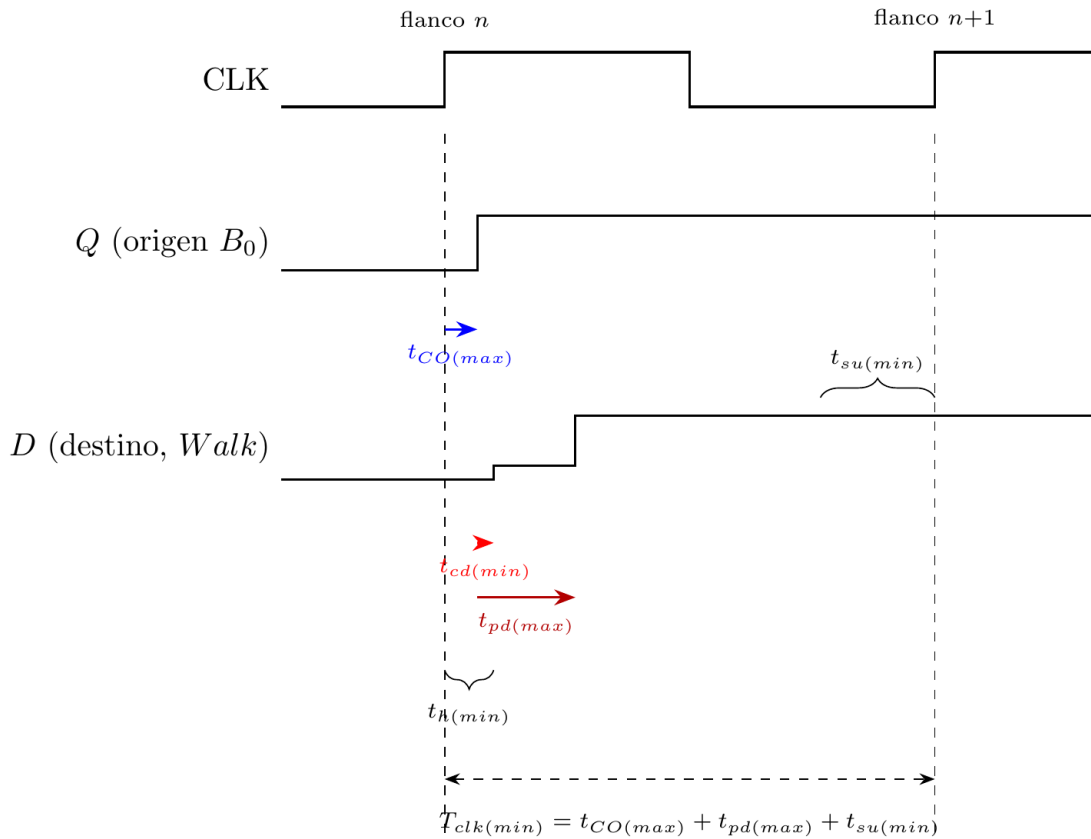


Figura 3. Secuencia de tiempo del tramo: en el flanco  $n$ ,  $Q$  cambia tras  $t_{CO(max)}$ . La señal se propaga entre  $t_{cd(min)}$  (usado para hold) y  $t_{pd(max)}$  (usado para setup).  $D$  debe estabilizarse con margen de  $t_{su(min)}$  antes del flanco  $n+1$ , y mantenerse estable  $t_{hd(min)}$  después del flanco  $n$ .

## 6. Cálculo de Contamination Delay (para el análisis de Hold)

El contamination delay (tcd) es el tiempo mínimo que tarda la señal en empezar a cambiar tras la ruta combinacional — el escenario más rápido posible, usando los valores MIN de cada compuerta, calculado sobre la ruta más corta:

$$tcd(comb) = tAND3(min) + tOR2(min) = 10 + 10 = 20 \text{ ns}$$

## 7. Cálculo de Setup y Frecuencia Máxima

$$Tclk(min) = tCO(max) + tpd(comb,max) + tsu(min)$$

$$\text{Retardo de la ruta crítica: } tpd(comb,max) = tNOT(max) + tAND4(max) + tOR2(max) = 24 + 28 + 25 = 77 \text{ ns}$$

$$\text{Período mínimo de reloj: } Tclk(min) = 44 + 77 + 25 = 146 \text{ ns}$$

$$Fc(max) = 1 / 146 \text{ ns} \approx 6.85 \text{ MHz}$$

**Resultado: el circuito puede operar de forma confiable hasta aproximadamente 6.85 MHz sin violar la restricción de setup.**

## 8. Cumplimiento de la restricción de Hold

$$tCO(min) + tcd(comb) \geq th(min)$$

$$20 \text{ ns} + 20 \text{ ns} = 40 \text{ ns} \geq th(min) = 0 \text{ ns}$$

**Resultado: 40 ns ≥ 0 ns — la restricción de hold se cumple con amplio margen (40 ns de holgura), sin necesidad de buffers adicionales.**

## 9. Conclusión — por qué se hizo así, y por qué se obtienen estos tiempos

**Por qué se eligió la familia 74HC: es una familia CMOS de propósito general, ampliamente documentada, disponible en Digikey en las cinco funciones lógicas exactas que necesita el circuito (flip-flop D, inversor, AND-3, AND-4, OR-2), lo que permite mapear la implementación de Logisim a piezas físicas reales sin combinar compuertas de dos entradas para simular las de más entradas.**

Por qué se usó  $VCC=4.5V$  y el rango extendido de temperatura: una fuente real de 5V nunca es perfectamente 5.000V — tiene tolerancia típica de  $\pm 10\%$ , así que el diseño debe garantizar funcionamiento incluso en el extremo inferior (4.5V), que es donde los fabricantes reportan los tiempos más lentos. De igual forma, se usó el peor caso de temperatura ( $-40^{\circ}C$  a  $85^{\circ}C$ ) en vez del valor típico a  $25^{\circ}C$ , porque un diseño de ingeniería debe garantizar el funcionamiento en todo el rango de operación esperado.

Por qué la ruta crítica es  $NOT \rightarrow AND-4 \rightarrow OR-2$  y no la otra: aunque ambos términos de la ecuación de Walk terminan en la misma compuerta OR, el segundo término necesita invertir dos señales antes de llegar a la AND, agregando una etapa completa de retardo adicional. La frecuencia máxima de un circuito siempre está limitada por su camino más lento, no por el promedio.

Por qué el hold se cumple tan fácilmente: la familia 74HC está diseñada internamente para que  $t_h(\min)=0$  ns — el fabricante garantiza que el dato no necesita permanecer estable ni un instante después del flanco de reloj. Como cualquier ruta real siempre tiene algún retardo mayor a cero, la condición de hold nunca representa un riesgo real en este diseño, a diferencia del setup.

En resumen: el cuello de botella de este sistema es la restricción de setup a través de la ruta combinacional más larga, y no la restricción de hold. La frecuencia máxima resultante,  $\approx 6.85$  MHz, es la velocidad más alta a la que se le puede pedir al reloj de la FSM que corra sin arriesgar errores de captura en el flip-flop de destino.